Tzu-Chieh Tai

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Education

University of Michigan Sept. 2025 – Jun. 2027 (expected)
M.S. in Robotics *Ann Arbor, MI*
GPA: 4.0/4.0
Courses: Deep Learning for Robotics, Mobile Robots, Self-Driving Cars

National Tsing Hua University Sept. 2020 – Jun. 2024
B.S. in Interdisciplinary Program of Engineering (ME and EE); GPA: 3.95/4.3 (Rank: 3/36) *Hsinchu, Taiwan*
GPA: 3.95/4.3 (Rank: 3/36)
Courses: Linear Algebra, Control System, Machine Learning, Robot Navigation and Exploration

Technical Skills

Robotics: SLAM(Kalman Filter, factor graph), 3D Reconstruction(OpenMVS, SfM, Trimesh), Mapping and Localization(Particle Filter), ROS, ROS2, Mobile Robots, Computer Vision(VIO, Yolo, CNN, Point Cloud, 3DGS, NeRF), Machine Learning, Sensor Integration, Deep Learning, Semantic Segmentation, Motion Planning, Sensor Calibration, Simulation(Pybullet, Gazebo)

Programming: C/C++ (GTSAM, Ceres, PCL, Eigen, OpenCV, Lie Group), Python (PyTorch, NumPy), OpenCV, Git, COLMAP, Open3D, Linux/Ubuntu, CMake, Docker, AWS

Engineering: LiDAR, Camera, IMU, CAD (Inventor, AutoCAD), Arduino, 8051, Microcontroller, Unreal Engine

Work Experience

Autonomous Systems Engineer Intern Jun. 2026 – Present
Tiami Networks *Sacramento, CA*

- Designed and implemented obstacle avoidance and person-tracking motion planning on the Skydio 2 drone, achieving a **90%** success rate in a 20m × 40m forested area, using only a single low-resolution RGB camera under high-latency conditions with TensorRT acceleration achieving 15 FPS in real time
- Developed and validated a vehicle-tracking algorithm for the Skydio 2 drone in urban environment, achieving zero tracking loss for vehicles traveling at up to 12 km/h across an average coverage area of 1,500 m^2
- Calibrated Depth Anything V2 metric outdoor model to design a reactive obstacle-avoidance planner integrated with OSNet for person Re-ID, achieving **zero collisions** and **zero tracking loss** across all test flights

Computer Vision & LiDAR Research Intern Mar. 2026 – Jun. 2026
Philo Homes *Ann Arbor, MI*

- Improved reconstruction quality by **30%** by developing hole detection algorithm from scratch, enabling real-time iOS hole detection and user guidance prompts via UI
- Reduced offline depth rendering time by **90%** (from 5 minutes to 30 seconds) using Nvdiffrast and vectorization, benchmarked unprojection, TSDF, and ray tracing methods for 3D reconstruction quality
- Engineered an end-to-end, full-stack 3D indoor reconstruction pipeline, spanning ARKit LiDAR capture and cloud-based processing with COLMAP, OpenMVS, AWS ECS, Step Functions, FastAPI, and Lambda

Research Assistant Oct. 2025 – Jun. 2026
MAVRIC Lab, University of Michigan *Ann Arbor, MI*

- Achieved **90%** accuracy by implementing a transformer-based semantic segmentation model
- Built up the pipeline for sensor fusion combining semantic information from model and distance measurements from LiDAR on UGV in Unreal Engine
- Building up data collection pipeline and LiDAR simulation from scratch in Unreal Engine to collect mask and raw RGB images for training models and corresponding raw LiDAR measurement

Research Assistant Mar. 2026 – May 2026
FROG Lab, University of Michigan *Ann Arbor, MI*

- Accelerated training by **2.2x** by implementing hard sampling and vectorizing sampling process on a 10k+ underwater dataset for structure reconstruction without loss of reconstruction accuracy

- Developed and optimized SIREN (NeRF-based) models for high-fidelity underwater scene reconstruction

Robotic Arm Software Intern Oct. 2023 – Oct. 2024

Gyro Systems Inc. *Hsinchu, Taiwan*

- Implemented reinforcement learning (TD3) control with digital twin (PyBullet) to optimize robotic arm's trajectories on AMR, resulting in improved motion efficiency during pick-and-place tasks
- Led integration of a collaborative robot arm with MoveIt! via ROS, developing a digital twin in C++ to enable comprehensive scenario testing and optimize execution accuracy and reliability
- Identified and resolved obstacle avoidance challenges in simulation by engineering motion planning strategies

Undergraduate Researcher Aug. 2022 – Jun. 2023

Reliable Computing Lab (Prof. Hsi-Pin Ma), National Tsing Hua University *Hsinchu, Taiwan*

- Achieved over **90%** autonomous space exploration coverage in unknown environments by implementing k-means, sensor fusion, and real-time navigation stacks in Python
- Integrated 2D/3D sensor inputs to enhance disaster area situation awareness and developed a distributed processing framework in ROS to support real-time casualty detection and autonomous exploration

Projects

Robust Visual Inertia Odometry from scratch Aug. 2026 – Present

- Developed visual intertia odometry pipeline for Luxonis OAK-D camera from scratch, levaraging sensor synchonization, multithreading, and preintegration to achieve robust real-time performance.

Autonomous Mobile Robot Exploration with LiDAR using MBot Oct. 2025 – Dec. 2025

- Reduced occupancy grid mapping error by **50%** (from 10 cm to <5 cm) by engineering a C++ thread-safe scan buffer in ROS2
- Effectively synchronized asynchronous LiDAR streams, eliminating data drops and ensuring high-fidelity data for real-time SLAM and scan matching
- Developed a custom scissor lift with a **2.8x extension ratio** (5 cm to 14 cm) for steady and balanced lifting of blocks

Uncertainty-Aware NeRF-SLAM Jan. 2026 – Apr. 2026

- Reduced ATE Mean by **28%** (0.745 → 0.537 cm) and ATE RMSE by **38%** (1.10 → 0.683 cm) on Replica dataset by integrating entropy-based uncertainty with covisibility for keyframe selection in NeRF-SLAM
- Recovered runtime overhead from 48% slowdown to 23% via fully vectorized batch uncertainty computation across all keyframes in a single forward pass
- Implemented multi-modal uncertainty combining geometry, depth, and semantic signals weighted by covisibility score for more informative NeRF updates

Multi-modal Visual Place Recognition and 3D Scene Understanding Jan. 2024 – Feb. 2024

- Achieved a **90% accuracy** in Recall@1 on HPointLoc and KITTI datasets by leveraging multi-modal representations from semantics and monocular depth for robust place recognition
- Developed a geometry-aware feature aggregation pipeline integrating DinoV2, semantic embeddings, and monocular depth features from Depth Anything V2 to enhance spatial consistency and large-scale scene retrieval
- Incorporated structure-aware representations inspired by Structure-from-Motion and 3D reconstruction pipelines to improve viewpoint invariance under illumination and seasonal changes

RGB-D Camera Perception Pipeline for 6-Axis Robotic Arm Aug. 2025 – Oct. 2025

- Led cross-functional integration of kinematics, calibration, and coordinate transformation components in Python and ROS2
- Achieved consistent **five-layer block stacking with under 10 mm error** by engineering a computer vision-based detection algorithm and optimizing closed-loop robotic motion control
- Overcame a computation stall in depth image projection, achieving precise real-time contour detection